

Title

Safety, tolerability, and preliminary efficacy of biomechanical foot interventions in individuals with knee osteoarthritis and foot pain: a pilot randomized feasibility study

Abstract

Background: Biomechanical interventions targeting the lower extremity can reduce knee joint loading; however, effects on concurrent foot symptoms are unclear. This pilot feasibility study examined angled foot inserts with and without arch support in patients with both knee osteoarthritis and foot pain. **Methods:** 30 adults with radiographically-confirmed knee osteoarthritis and foot pain were randomized to receive either 6-degree angled inserts alone (n=15) or angled inserts integrated with contoured foot and arch support (n=15) for 12 weeks. Feasibility measures included recruitment/retention rates, daily wear compliance, comfort ratings, and adverse event frequencies. Clinical measures included foot function indices, knee symptoms questionnaires, and motion capture analysis of lower limb biomechanics. **Results:** All feasibility criteria were successfully met with no serious adverse events reported. Participants rated insert comfort as moderate (mean 7.0/10), and 83% of weeks met the target 30-minute daily wear threshold. Both groups showed consistent improvements in knee pain and function scores, knee adduction moment reduction, and ankle stability measures. No statistical comparisons were conducted due to pilot design. **Conclusions:** Both insert types appear safe and feasible in this population; larger confirmation studies are warranted.